

Pull Expired Lots (FEFO) SOP

One episode clears one bin. It runs from a bin holding six dated items, a bare check mat, an empty quarantine tote, a blank log card and an old bin card in the bin's card holder, to the two expired items lying in the quarantine tote with their lot labels on the log card, the four in date items back in the bin, and a new bin card in the holder showing the earliest date left in the bin.

The order never changes: **check the date on every item, pull the expired ones to the tote and log their lot numbers, put the in date items back in the bin, then change the bin card.** No item is pulled before every date has been read. No item is logged after it is in the tote. The bin card is changed last, and it is never changed before the bin is full again.

One gripper holds and the other gripper works. The holding gripper is closed and empty and presses the object down so it stays still. The working gripper picks up, carries, presses and places. Nothing on this station needs both grippers at once. Nothing is ever passed from one gripper to the other, only one object is held at a time, and no grasp is changed while an object is off the surface.

The bin and the check mat sit on the midline and are shared, so each gripper meets them from its own side. Nothing is carried across the table to the far side, and no carry passes over the bin, over the tote or over the log card.

The counts are fixed: **one bin, six items, six date reads, two expired items, two lot labels and one new bin card.** There is no second pass over the bin and no second read of an item. A date is never read again once the item is on its line. An item, a label or a card that lands outside its place is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 90 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the today card, the bin card rack, the used card spot, the bin, the check mat, the log card and the quarantine tote.
3. The date on the today card is readable in the frame for the whole episode. The date label on

an item is readable while the item is held up, and the printed date on a bin card is readable before the card goes into the holder.

4. Both arms are at home with their grippers open.
5. The tabletop is dry and clear of anything but the items in the Materials checklist.
6. Each arm reaches the today card, the bin card rack, the used card spot, the bin, the whole check mat, the log card and the quarantine tote without extending to a joint limit or folding in on itself. Bring the mat and the tote forward until this is true.
7. Each arm reaches the bin and the check mat from its own side of the midline. Neither has to lean over the other to work them.
8. Neither arm needs the far side of the table for anything. The paths from the bin forward to the mat, from the mat to the log card, from the mat to the tote, and from the card rack forward to the bin are clear. Carry one item forward by hand during setup: it must reach the mat, the log card and the tote without passing over the bin. Widen the gaps until it does.

Materials checklist

1. The check mat lies flat at the front center of the table, straddling the midline. It grips the table. Draw a hand across it during setup: if it slides, bunches or lifts, weight it or replace it with a rubber mat that stays put. Nothing pins the mat during the episode.
2. The mat is marked with two lines across it, a keep line at the back and a pull line at the front. Each line is long enough to hold six items side by side with a gap between them. The mat is bare at the start.
3. One bin stands at the bin spot, open at the top and shallow enough that every item in it is visible. It holds six items and nothing else. It stands still on the table and does not lean.
4. The six items are the same shape and size. Each one is dry, light enough for one gripper to carry level, and square enough for the gripper to close on it and hold it level.
5. Each item carries one printed date label on one face, showing one date and nothing else. The date is large enough to read on camera. A smudged, torn or handwritten date label is not usable.
6. Each item carries one peel off lot label beside its date label, printed with the lot number and lifted at one corner so a gripper can peel it. The lot label sticks down on the log card in one press. A label that curls back on itself is replaced.
7. Two of the six items are expired and four are in date. The four in date items carry four different dates, so one of them is clearly the earliest.
8. One today card stands upright at the back left, facing the camera, printed with one date. This is the date every item is checked against. The same card is used for the whole episode.
9. One log card lies flat on the log card spot. It is ruled into six blank rows, each row wide enough for one lot label with a clear edge all the way round. No row has a label on it at the start.

10. One quarantine tote stands empty at the front right, marked QUARANTINE, shallow enough that the items inside are visible, and wide enough that an item goes in without being pushed.
11. One card holder is fixed to the front of the bin, facing the camera. It holds one card at a time, upright and readable, and a card slides in and out of it without the bin being lifted or held down.
12. One bin card rack stands at the back right, holding one card for each date carried by the six items, standing in a row and readable. The old bin card sits in the bin's card holder. The used card spot beside the rack is bare.

Workspace layout

Three rows: the today card and the bin card rack at the back, the bin in the middle, the check mat, the log card and the quarantine tote at the front. Everything travels forward toward the mat.

Back row, left to right:

1. Today card, back left, standing upright and facing the camera.
2. Bin card rack, back right, the new date cards standing in a row.
3. Used card spot, the strip of bare table beside the rack. Empty at the start, one old card lying flat on it at the end.

Middle row:

4. Bin spot, center on the midline, the bin standing open with its card holder facing the front.
Shared: one gripper holds it, the other works it, and each meets it from its own side.

Front row, left to right:

5. Log card spot, front left beside the mat, the log card lying flat with its rows facing up.
6. Check mat, front center, straddling the midline, the keep line across its back and the pull line across its front. Bare at the start and bare again at the end.
7. Quarantine tote, front right beside the mat. Empty at the start, two expired items in it at the end.

A clear gap of at least a gripper's width sits between the bin and the mat, between the mat and the log card, and between the mat and the tote, so an object lifts straight up and travels forward without touching anything. No carry passes over the bin, over the tote or over the log card.

Vocabulary

- **Hold:** a closed empty gripper pressed onto an object to keep it still. A hold never lifts, carries or moves what it holds.
- **Work:** the other gripper picking up, carrying, pressing or placing. One gripper holds and the

other works, and the two never trade jobs part way through a step.

- **Home spot:** the spot an object is laid on and lies flat on before the next step starts: the log card spot, the used card spot, the check mat, the bin, the quarantine tote. Nothing is left on bare tabletop or in a gripper between steps.
- **Release point:** the gripper opens over a line on the check mat, over the log card, inside the mouth of the bin, inside the mouth of the quarantine tote, or over the used card spot. Never over the today card and never over open tabletop.
- **Stable placement:** the object lies still after release, flat on its surface, and does not rock, tip, slide or lean on anything.
- **Item:** one of the six dated units in the bin. Every item is checked once and ends up either back in the bin or in the quarantine tote.
- **Date label:** the printed label on the item that gives its date. It is read, never peeled.
- **Today card:** the standing card at the back that gives today's date. Every item date is compared with this one date.
- **Date read:** the date label turned to the camera and held still for 1 second, with the today card in the same frame. The item is kept low over the mat.
- **In date:** the date on the item is the same as the date on the today card, or later than it. An in date item goes on the keep line.
- **Expired:** the date on the item is earlier than the date on the today card. An expired item goes on the pull line.
- **Keep line:** the line across the back of the mat. In date items lie on it, side by side, with a gap between them.
- **Pull line:** the line across the front of the mat. Expired items lie on it, side by side, with a gap between them.
- **Lot label:** the peel off label on the item printed with its lot number. It is taken by its lifted corner, never by its sticky face.
- **Log card:** the ruled card the lot labels are pressed onto. It stays flat on its spot the whole episode.
- **Logged:** one lot label lying flat in one open row of the log card, the lot number readable, no part of the label over a rule line, and no bubble under it.
- **Open row:** a row of the log card with no label on it. Rows are filled from the top down, one label per row.
- **Quarantine tote:** the marked tote the expired items go into. Nothing else ever goes in it.
- **Earliest date:** the soonest date left on the keep line once every item has been read. This is the date the new bin card shows.
- **Bin card:** the card in the holder on the front of the bin. It shows one date, and one card only is ever in the holder.
- **Relabelled bin:** the old card out and lying on the used card spot, and the new card seated

upright in the holder with its date readable.

- **Spill:** an item, a lot label or a card that ends up outside its place. It is left where it falls and cleared in the reset, never on camera.
- **Quarter turn:** the most any object is tilted in a gripper. Turning a date label up and laying an item back down both stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Check the date on every item

Goal: the bin is empty, all six items lie on the mat, the in date ones on the keep line and the expired ones on the pull line, and the earliest date has been chosen.

Every item is read once, on its way out of the bin. An item is never read again once it is on a line, and the two lines are never mixed.

1.1 Take one item out of the bin and read its date

- With one gripper closed and empty, come down on the rim of the bin and press it against the table so it stays still. Take this hold **before the other gripper goes into the bin**, and keep it **until the bin is empty**.
- With the other gripper, close on one item, lift it straight up clear of the bin, and carry it forward over the mat.
- Turn the item **no more than a quarter turn** so the date label faces the camera, and keep it low over the mat.
- Hold it still for **1 second** with the today card in the same frame, so the two dates are seen together.

The item is kept low over the mat the whole time. An item turned over in the air is out of frame for part of the turn, and a date that is never framed cannot be checked.

Check: the date label was in the camera frame and still for 1 second. If it was hidden or blurred, show it once more. Make no more than two extra reads for one item.

1.2 Lay the item on its line

- Compare the two dates. The item is **expired** if its date is **earlier** than the date on the today card. It is **in date** if its date is the **same or later**.
- Lower an in date item onto the **keep line** and an expired item onto the **pull line**, date label facing up, and release once it lies still.
- Leave a clear gap between the items on a line. They do not touch, and none of them overhangs a mat edge.
- Repeat 1.1 and 1.2 until the bin is empty. **Six items in total**. The holding gripper releases

the bin and lifts clear only now.

Once an item is on a line it stays there until its own step moves it. It is not read again and it is not moved to the other line.

Check: the bin is empty, six items lie on the mat, every one of them lies fully on a line with its date label facing up, and no item touches another. If an item overhangs an edge or touches another, lift it clear and lay it down again. Make no more than one correction per item.

1.3 Choose the earliest date

- Read across the keep line and choose the **earliest date** on it. This is the date the new bin card will show.
- Choose once. **The date is not changed later in the episode.**

Check: one date has been chosen and it is the soonest date on the keep line. If two items on the keep line show the same earliest date, that date is still the choice.

Step 2: Pull the expired items and log their lot numbers

Goal: the pull line is bare, both expired items lie in the quarantine tote, and both lot labels lie in the log card, one to a row.

The lot number is logged while the item is still on the mat. An item in the tote is never worked on again.

2.1 Log the lot number

- With one gripper closed and empty, come down on one expired item on the pull line and press it against the mat. Take this hold **before the label is peeled**, and keep it **until the label is on the log card**.
- With the other gripper, take the lot label **by its lifted corner** and peel it off the item in one draw, sticky side down.
- Carry it to the log card, lower it into the **top open row**, and press it down once.
- Release the label and lift clear.

Never take a lot label by its sticky face. A label taken flat on its sticky side folds onto itself and cannot be laid down.

Check: the label lies flat inside one row, the lot number is readable, no part of it sits over a rule line, and there is no bubble under it. If a corner is lifted, press the same label down again. Make no more than one correction. Never lay a second label on top of the first, and never put two labels in one row.

2.2 Put the item in the tote

- With one gripper, close on the item, lift it straight up clear of the mat, and carry it level to the quarantine tote.
- Lower it inside the mouth of the tote until it lies on the bottom, then open the gripper and lift clear.
- Repeat 2.1 and 2.2 for the second expired item. **Two items in total.**

The carry goes straight from the mat to the tote and passes over nothing else. An item is never dropped in from above the tote rim.

Check: the pull line is bare, both expired items lie in the tote, and two lot labels lie in the log card, one to a row. If an item ends up on the tote rim or half out, lift it clear and lower it in again. Make no more than one correction per item.

Step 3: Put the in date items back in the bin

Goal: the keep line is bare, all four in date items are back in the bin, and the mat is bare.

One item per carry, taken from the keep line in the order they lie, working from one end to the other. No item goes back in the bin until its date has been read.

- With one gripper closed and empty, come down on the rim of the bin and press it against the table so it stays still. Take this hold **before the first item comes over**, and keep it **until the last item is in**.
- With the other gripper, close on one item on the keep line, lift it straight up clear of the mat, and carry it level back to the bin.
- Lower it inside the mouth of the bin until it lies on the bottom or on the items already there, then open the gripper and lift clear.
- Repeat until the keep line is bare. **Four items in total.**
- The holding gripper releases the bin and lifts clear only now.

Nothing is stacked above the bin rim, and no item is pushed down to make room. A bin that will not take its four items back is a reset error.

Check: the mat is bare, four items lie in the bin, all of them below the rim, and the bin still stands square on its spot. If an item sits on the rim or leans out, lift it clear and lower it in again. Make no more than one correction per item.

Step 4: Change the bin card

Goal: the old card lies flat on the used card spot and the new card stands in the holder showing the earliest date chosen in Step 1.3.

The card is changed last, once the bin holds only in date items. The bin is not lifted, tipped or slid while the card is changed.

4.1 Take the old card out

- With one gripper closed and empty, come down on the rim of the bin and press it against the table so it stays still.
- With the other gripper, close on the top edge of the old card and draw it straight up out of the holder.
- Carry it to the used card spot, lay it flat, release and lift clear.

Check: the holder is empty and the old card lies flat on the used card spot, not on the tabletop, not on the mat and not in the tote.

4.2 Put the new card in

- Take from the bin card rack the one card whose printed date is the **earliest date chosen in Step 1.3**, closing on its top edge and lifting it straight up out of the rack.
- Carry it to the holder, line its bottom edge up with the slot, and lower it in until it is seated and stands upright on its own.
- Release and lift clear. The gripper holding the bin releases and lifts clear only now.

One card in the holder, and one only. The new card is not slid in in front of the old one.

Check: one card stands in the holder, its date is readable in the frame, and it matches the earliest date on the items now in the bin. If the card leans, sits crooked or falls out, draw it out and seat it again. Make no more than one correction. If the wrong card went in, take it out, lay it back in the rack and take the right one, once.

Step 5: End the episode

- Confirm the mat: bare, both lines clear, nothing left on it.
- Confirm the bin: four in date items in it, all below the rim, and the bin square on its spot.
- Confirm the tote: two expired items in it and nothing else.
- Confirm the log card: two lot labels on it, one to a row, both readable, and the card flat on its spot.
- Confirm the cards: the new card upright in the holder showing the earliest date, and the old card flat on the used card spot.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return both arms home with their grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the two items out of the quarantine tote and take the four items out of the bin, and

leave the tote empty.

2. Peel the two lot labels off the log card, bin the used labels and the used log card, and lay a fresh log card flat on its spot with six blank rows facing up.
3. Put a fresh lot label on each item that lost one, printed with its lot number and lifted at one corner. A label that has been peeled once is not used again.
4. Put the six items back in the bin, two expired and four in date, the four in date ones carrying four different dates. Change the date labels or swap items in so the same set is not run twice in a row.
5. Take the card out of the holder, put the old card back in from the used card spot, and stand the rack cards in a row again, one for each date now in the bin. Leave the used card spot bare.
6. Stand the today card upright at the back left and check that its date still makes two items expired and four in date. Change the card if it does not.
7. Pick up any item, label or card that fell during the episode, and wipe the tabletop and the mat dry.
8. Confirm the mat lies flat and still grips the table, both lines on it are still clear to see, the bin and the tote are not cracked or leaning, and the card holder still holds a card upright.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** an item goes to the tote before every item has been read, an item goes back in the bin before every item has been read, a lot label is peeled after its item is in the tote, the bin card is changed before the in date items are back in the bin, or a step is skipped.

- **SOP rule broken:** Steps 1 to 4, check, pull and log, restock, relabel, in that order, with the log done while the item is still on the mat.
- **Coaching note:** read them all, pull and log, put the rest back, then change the card. The card is the last thing that moves.

Violation: Date not read in the frame

- **Visible cue:** an item is laid on a line while its date label has never been in the camera frame, the label is shown for less than 1 second, the label is shown while the item is moving, or the today card is out of frame while a date is shown.
- **SOP rule broken:** Step 1.1, the date label is turned to the camera and held still for 1 second with the today card in the same frame.
- **Coaching note:** one item at a time, still for a full second, with both dates in the shot. A date that was never framed cannot be checked.

Violation: Item put on the wrong line

- **Visible cue:** an item whose date is earlier than the today card date is laid on the keep line, an item whose date is the same or later is laid on the pull line, or an item is moved from one line to the other after it has been released.
- **SOP rule broken:** Step 1.2, expired goes on the pull line, in date goes on the keep line, and an item stays on its line once it is released.
- **Coaching note:** earlier than today is expired. Same day is still good. Decide before the gripper opens.

Violation: Item count wrong, or the bin read twice

- **Visible cue:** fewer or more than six items are taken out and read, an item is left in the bin unread, an item is lifted out and read a second time, or a second pass is made over the bin after it is empty.
- **SOP rule broken:** Steps 1.1 and 1.2, six items, one read each, one pass over the bin.
- **Coaching note:** six out, six read, six on the mat. A second look makes the episode unlearnable.

Violation: Bin not held, lifted, or knocked over

- **Visible cue:** a gripper is off the bin rim at any point while items are being lifted out or put back; the bin lifts, shifts or tips as an item is drawn out; or the bin leaves the table, falls on its side, or is knocked out of the bin spot.
- **SOP rule broken:** Steps 1.1, 3 and 4, one gripper holds the bin rim throughout, and the bin stays standing on the table.
- **Coaching note:** the holding gripper holds through the whole run, not just the first item. The bin is worked in place and is never picked up.

Violation: Lot label taken or pressed wrong

- **Visible cue:** a lot label is taken by its sticky face, is torn part way off and left on the item, folds onto itself, is pressed over a rule line, is pressed onto a row that already holds a label, is laid on top of another label, or ends up with the lot number face down or unreadable.
- **SOP rule broken:** Step 2.1, the label is taken by its lifted corner, peeled off in one draw, and pressed flat inside the top open row with its number readable.
- **Coaching note:** corner first, one clean peel, one row per label, filled from the top down.

Violation: Item pulled with no lot logged

- **Visible cue:** an expired item goes into the quarantine tote with its lot label still on it, or the episode ends with fewer than two labels on the log card.
- **SOP rule broken:** Step 2, every item that goes to the tote has its lot label logged first, two labels in total.
- **Coaching note:** log it, then move it. A pull with no lot number on the card cannot be traced.

Violation: Wrong item in the tote or in the bin

- **Visible cue:** an in date item is put in the quarantine tote, an expired item is put back in the bin, an item is lifted back out of the tote, or anything other than an expired item is put in the tote.
- **SOP rule broken:** Steps 2.2 and 3, the pull line goes to the tote, the keep line goes to the bin, and nothing comes back out of the tote.
- **Coaching note:** the tote takes expired stock only, and it is a one way trip.

Violation: Wrong bin card, or the date changed after it is chosen

- **Visible cue:** the card put in the holder shows a date that is not the earliest date on the items now in the bin, a card is put in and then swapped for another more than once, or the card of an item that went to the tote is used.
- **SOP rule broken:** Steps 1.3 and 4.2, the earliest date is chosen once from the keep line, and the card that goes in the holder shows that date.
- **Coaching note:** the card shows the soonest date still in the bin. Pick it once and stand by it.

Violation: Old card left in, or two cards in the holder

- **Visible cue:** the new card is put in while the old card is still in the holder, two cards stand in the holder, the old card is left on the tabletop, on the mat or in the tote, or the holder ends the episode empty.
- **SOP rule broken:** Steps 4.1 and 4.2, the old card comes out to the used card spot first, then one card goes in.
- **Coaching note:** out first, then in. One card in the holder, always.

Violation: Hold and work traded part way

- **Visible cue:** the gripper that was holding starts working, or the gripper that was working takes over the hold, part way through a step; or both grippers work at once with nothing holding the bin or the item down.
- **SOP rule broken:** Steps 1 to 4, one gripper holds and the other works, and the two never trade jobs part way through a step.
- **Coaching note:** whichever gripper takes the hold keeps it to the end of the step. Nothing here needs both at once.

Violation: Spill cleaned up

- **Visible cue:** after an item, a label or a card lands outside its place, a gripper is used to sweep, push, or pick it up.
- **SOP rule broken:** Steps 1 to 4, a spill is left where it falls and is cleared in the reset.
- **Coaching note:** cleanup motions are not part of the task. The spill is data, the cleanup is noise.

Violation: A carry passed over the bin, the tote or the log card

- **Visible cue:** an item is carried over the open bin on its way to the tote, over the tote on its way to the bin, or over the log card; or a card is carried over the bin mouth or over the tote.
- **SOP rule broken:** Steps 2.2, 3 and 4, each carry runs straight to its own spot and passes over nothing else.
- **Coaching note:** route around, not across. An open bin under a carry is where a wrong item ends up.

Violation: Bad pick, carry, or release

- **Visible cue:** an object moves from one gripper to the other without being set down first, both grippers hold an object at once, a gripper shifts or seats its hold again while the object is off the table, an item or a card slides along the table with no daylight under it, an item hangs tilted in the gripper because the grasp was taken off square, an item is dropped in from above a rim, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 4, one object at a time, lifted straight up clear of the surface before any horizontal motion, tilted no more than a quarter turn, lowered until it touches, and never passed between grippers.
- **Coaching note:** set it down to change the grip, and lift before the carry. Lower it in, do not drop it in.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** an item, a card, the bin, the tote or the mat is shoved, nudged, or slid into position instead of being lifted clear and placed again; an item in the bin is pressed down to

make room; or a failed check is retried more often than its step allows.

- **SOP rule broken:** Steps 1 to 4, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Object not returned to its home

- **Visible cue:** a gripper opens while an item is still above the bin rim or the tote rim; an item is left on the mat, on the tabletop or on the log card; the old card is left off the used card spot; or an object is still in a gripper when the next step begins.
- **SOP rule broken:** Steps 2.2, 3, 4.1 and 4.2, every object is lowered onto its own spot and lies or stands there before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, a gripper or a carried object strikes the bin, the tote, the today card, the card rack, the log card or an item on the mat and knocks, tips, slides, or displaces it; the two arms strike each other over the bin; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 4, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. On the bin, the working gripper moves and the holding gripper stays put.

Violation: Manipulation outside the camera frame

- **Visible cue:** a date read, a placement, a peel, a press or a card change happens partly or wholly outside the environment camera frame.
- **SOP rule broken:** Steps 1 to 4, every manipulation happens inside the camera frame.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item still on the mat, an expired item still in the bin, a lot label missing from the log card, the holder empty or holding the old card, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 5, confirm the mat, the bin, the tote, the log card and the cards, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do before the recording

ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside an arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective mat: one that slides, bunches or lifts under a correct carry, one whose lines cannot be told apart, or one that will not lie flat. Weight or replace it before the next episode.
- Defective bin: one too deep to see the items in, too narrow for an item to go in without being pushed, that will not stand still, or that is cracked or leaning.
- Defective tote: one too deep to see the items in, that is not marked QUARANTINE, or that will not stand still.
- Unreadable date label at episode start: smudged, torn, handwritten, or too small to read on camera.
- Defective lot label: one with no lifted corner, one that tears as it is peeled, one that curls back on itself, or one that will not stick to the log card in one press.
- Defective log card: one with no open row left, one whose rows are too small for a label, or one that will not lie flat.
- Defective card holder: one that will not hold a card upright, or that a card cannot be drawn out of without the bin moving.
- Today card missing, fallen over, or facing away from the camera at episode start.
- The bin card rack holds no card for the earliest date on the keep line, or the bin holds a number of items other than six at episode start. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Hold the bin steady on its spot
2. Lift one item out of the bin
3. Turn the item to show its date label
4. Lay one item on the keep line
5. Lay one item on the pull line
6. Hold one item steady on the mat
7. Peel one lot label off an item

8. Press one lot label into a row of the log card
9. Lift one expired item from the pull line
10. Put one expired item in the quarantine tote
11. Lift one in date item from the keep line
12. Put one in date item back in the bin
13. Draw the old bin card out of the holder
14. Lay the old bin card on the used card spot
15. Take one new bin card from the rack
16. Seat the new bin card in the holder
17. Return both arms home and end the episode